18. Implement Perceptron based IRIS classification.

```
IDLE Shell 3.13.5 - C:/Users/saimo/OneDrive/ドキュメント/Desktop/ML Programs/18.py (3.13.5)
le  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: C:/Users/saimo/OneDrive/ドキュメント/Desktop/18th program.py ======
=====

Epoch: 1
Expected: 0 Predicted: 1 Error: -1
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 1 Predicted: 0 Error: 1
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0

Epoch: 2
Expected: 0 Predicted: 1 Error: -1
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 1 Predicted: 0 Error: 1
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0

Epoch: 3
Expected: 0 Predicted: 1 Error: -1
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 1 Predicted: 0 Error: 1
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0

Epoch: 4
Expected: 0 Predicted: 1 Error: -1
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0

Epoch: 5
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 0 Predicted: 0 Error: 0
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0
Expected: 1 Predicted: 1 Error: 0
                                                                    Ln: 92  Col: 0
```

```
18th program.py - C:/Users/saimo/OneDrive/ドキュメント/Desktop/18th program.py (3.13.5)
File  Edit  Format  Run  Options  Window  Help

weights = [0.0, 0.0, 0.0, 0.0]
bias = 0.0
learning_rate = 0.1
epochs = 10


def step_function(x):
    return 1 if x >= 0 else 0
for epoch in range(epochs):
    print("\nEpoch:", epoch+1)

    for row in dataset:
        x = row[:-1]
        y = row[-1]

        z = sum(w*x[i] for i, w in enumerate(weights)) + bias

        y_pred = step_function(z)


        error = y - y_pred

        for i in range(len(weights)):
            weights[i] += learning_rate * error * x[i]

        bias += learning_rate * error

        print("Expected:", y, "Predicted:", y_pred, "Error:", error)

def predict(sample):
    z = sum(weights[i]*sample[i] for i in range(len(weights))) + bias
    return step_function(z)


test_sample = [5.0, 3.2, 1.5, 0.2]

result = predict(test_sample)

print("\nFinal Weights:", weights)
print("Final Bias:", bias)

print("\nTest Sample:", test_sample)
print("Predicted Class:", result)
print("0 = Setosa, 1 = Versicolor")
                                                                    Ln: 45  Col: 0
```